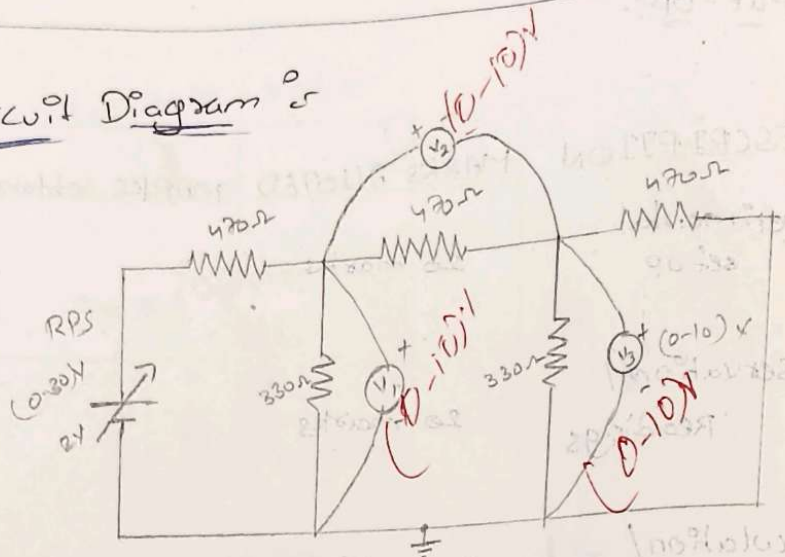
Circuit Diagram :



EXP No:- 03
Date :- 6/2/26

DETERMINATION OF VOLTAGE IN CIRCUIT USING NODAL ANALYSIS

Aim :-

To determine the voltage in the circuit using nodal analysis both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED :-

S.No	Apparatus	specification	Quantity
1.	Regulated power supply	(0-30)V	1
2.	Multimeter	–	1
3	Resistors	470 Ω, 330 Ω	3, 2
4	Bread board	–	1

THEORY :-

In electric circuits analysis, nodal analysis, node-voltage, analysis or the branch current method is a method of determining the voltage (potential difference) b/w "nodes" (points where elements or branches connect) in an electrical circuit in terms of branch currents.

TABULAR COLUMN :-

Parameters	Theoretical	Practical
V_1	0.63	0.66
V_2	0.18	0.19
V_3	0.45	0.47

CALCULATIONS :-

Nodal analysis

At node A:-

By KCL:-

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$\frac{33V_A - 66 + 47V_A + 33V_A - 33V_B}{15570} = 0$$

$$113V_A - 33V_B = 66 \quad (1)$$

At node B

By KCL:

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$33V_B - 33V_A + 47V_B + 33V_B = 0$$

PROCEDURE :-

1) Give connections as per circuit diagram.

2) Switch ON the supply - vary the RPS and set a particular input voltage.

3) Note down the readings of ammeters and ~~voltage~~ voltmeters and tabulate them.

4) Vary the RPS for different input voltages and note down the readings of all the meters.

5) Reduce the RPS to its minimum value and switch OFF the supply.

6) Using the tabulated values verify kirchoff's law practically and verify it theoretically.

$$113V_B - 33V_A = 0$$

$$113V_B = 33V_A$$

$$V_A = \frac{113V_B}{33} \quad - (2)$$

sub (2) in (1)

$$113\left(113\frac{V_B}{33}\right) - 33V_B = 66$$

$$\frac{12769}{33}V_B - \frac{1089}{33}V_B = 66$$

$$V_B = \frac{2178}{11680}$$

$$\boxed{V_B = 0.186\,V}$$

$$V_A = \frac{113}{33}V_B = \frac{113 \times 0.18}{33}$$

$$\boxed{V_A = 0.638\,V}$$

$$V_1 = V_A = 0.63\,V$$

$$V_2 = V_B = 0.18\,V$$

$$V_3 = V_A - V_B = 0.63 - 0.18$$

$$V_3 = 0.45\,V$$

Marks split-up:

S.No	Description	Marks allotted	Marks obtained
1.	Experimental setup	20 marks	20
2.	observation/ readings	20	20
3.	calculation/ result	20	20
4.	viva	20	~~20~~ 18
5.	Record	20	18
6.	Total	100	96

Result:

Thus, Nodal analysis is verified practically and theoretically. The resultant voltages for 2 V supply are

a) The voltage V1 is 0.63 V.

b) The voltage V2 is 0.18 V.

c) The voltage V3 is 0.45 V.